

# The performance of model based option trading strategies

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**Abstract** This paper analyzes returns to trading strategies in options markets that exploit information given by a theoretical asset pricing model. We examine trading strategies in which a positive portfolio weight is assigned to assets which market prices exceed the price of a theoretical asset pricing model. We investigate portfolio rules which mimic standard mean-variance analysis is used to construct optimal model based portfolio weights. In essence, these portfolio rules allow estimation risk, as well as price risk to be approximately hedged. An empirical exercise shows that the portfolio rules give out-of-sample Sharpe ratios exceeding unity for S&P 500 options. Portfolio returns have no discernible correlation with systematic risk factors, which is troubling for traditional risk based asset pricing explanations.

**Keywords** Dynamic trading · Options returns · Stochastic volatility · Mean-variance

**JEL Classification** G1 · G13 · G17

## 1 Introduction

Asset pricing models more often than not produce theoretical prices which deviate from prices observed in the market. These deviations, commonly termed ‘pricing errors’, can be given different interpretations. Typically researchers look at the size of the errors to assess the success of the model. The conventional wisdom is that smaller errors implies a better model. This view prevails throughout the applied finance literature. In the term structure literature, for example, models with latent factors

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produce much smaller pricing errors than, say, structural equilibrium models based on macro factors. Another example is the cross-sectional expected returns literature, where the reduced form models typically produce much smaller pricing errors than more structural equilibrium based models. Thus, in both strands of the literature it appears almost uniformly accepted that small pricing errors is the metric on which a model's success should be judged. This paper argues that pricing errors can be interpreted differently. In fact, a model's success may be completely inverse to its pricing error. This is true if the model is correct, and the market is not because in this case, pricing-errors can be exploited to form profitable trading strategies. In other words, if an asset market is inefficient and admits systematic mis-pricing, and a particular theoretical asset pricing model provides the correct prices, the deviations between the market and model prices (pricing errors) could be exploited to form profitable, low risk trading strategies by a rational speculator.

The traditional view of pricing errors is that the smaller they are, the more successful the model is. This is a reasonable statistical decision rule under the assumption that markets are perfectly rational and efficient. A statistical comparison of two models based on, say, mean squared pricing errors implies a quadratic loss function and implicitly assumes rational, efficient markets. That is, the only way to reasonably conclude that model A is better than B because A produces smaller pricing errors than B, is by assuming that markets are perfectly rationally pricing the assets. By contrast, this paper studies a different loss function based on expected utility from trade (EUT). We argue that if markets are not fully efficient, additional information about a models ability to generate abnormal returns provides an alternative and useful diagnostic for model comparison.

If we take the view that pricing errors may occur because the market prices are not correct, we may then ask to what extent deviations from rational no-arbitrage or equilibrium prices may be exploited. This is akin to choosing a statistical loss function to evaluate models under the alternative assumption of imperfect markets. In specifying a EUT criterion, we specifically assume a mean-variance investor who maximizes expected utility with a given planning horizon. Solving the mean-variance problem with  $N$  assets under consideration requires an  $N$  vector of expected returns, as well as the  $N \times N$  covariance matrix. We choose both based on the asset pricing model: Expected rates of return are computed using the difference between the model and market prices (pricing errors).

In exploring these ideas empirically, the paper presents evidence from options markets data. Of course the idea of EUT can be applied to any asset market, but options markets present a near perfect test-bed. There are two reasons for this. First, options pricing models produce theoretical prices without assumptions that are hard to verify. This contrasts, for example, equity pricing models which produce expected returns, but not necessarily prices. Second, in options markets uncertainty is resolved at the expiration of the contract. This is important because if uncertainty is not resolved and the bet is settled, one might argue that markets are right even when they're wrong. Consider the stock market: if a stock is objectively mispriced today, it may also, and most likely will be, mispriced in the future. Thus, betting against the market is a losing proposition, even if the speculator is "right" by some seemingly objective measure.

In examining the EUT, we formulate an optimal portfolio problem in the following way. First, expected returns are taken to be the difference between model and market prices. Second, based upon a particular model, we compute the covariance matrix of returns on the respective options contracts. The covariance is computed by integrating with respect to the two random state-variables that affect options prices—the future stock price and the volatility of the underlying stock. In this case, the mean-variance criterion maximizes the expected returns from trade subject to the minimum attainable variance. To gain some intuition for how this procedure works, consider the case of two call options with adjacent strikes, and assume that the model produces prices which exceed the market prices. If we were to simply maximize expected return, we would take positive positions in both contracts. Because the payoffs on the contracts are almost perfectly positively correlated, buying both contracts while disregarding the systematic risks incurred by essentially putting on the same bet is clearly bad from the point of view of a risk averse investor because the position is almost as risky as just buying a single contract. In this situation, it may prove optimal to buy the highest expected return contract, and sell the other one, fully expecting to take a loss on the second position because this provides almost a perfect hedge. We outline the optimal portfolio approach in detail below.

We examine the EUT approach empirically using data from S&P 500 options markets. The benchmark models used are affine jump diffusion models which are well understood, and well studied in the empirical options pricing literature. It is a well known fact that options on the S&P index seemingly sell at very high market prices. Two empirical observations support this notion. First, implied volatility—the volatility which equates market prices to the theoretical price, seems systematically higher than measures of volatility based on the variability of the underlying returns. This so called volatility premium suggests an ex-ante negative premium long options positions. Second, a number of papers suggest that issuers of options earn very high rates of returns, seemingly while taking on low risk. [Coval and Shumway \(2001\)](#) document annualized Sharpe ratios of about one to S&P 500 options writers over the '87–'96 sample period. This is about twice what investors would get from a buy and hold position in the underlying S&P 500. [Jones \(2006\)](#) reports Sharpe ratios that from about 0.3 to 1.07 for various strategies. [Santa-Clara and Saretto \(2009\)](#) examine the impact of margins on option returns and find that the negative returns persist after controlling for margin. The implications for rational asset pricing models of seemingly high returns to options writers have also been examined in [Driessen and Maenhout \(2007\)](#). They apply the GMM framework of [Ait-Sahalia and Brandt \(2001\)](#) to non-parametrically estimate utility functions of the representative investor from options returns. Their analysis reveal that not even a relative risk aversion coefficient of 50 is sufficient for an CRRA investor to demand the kind of average return premiums found in options markets. They conclude that their evidence is consistent with large risk premiums for volatility and jump risks. [Bakshi and Kapadia \(2003\)](#) show that gains to writing options are significantly increasing in the level of market volatility. This is consistent with the affine jump diffusion models considered here which have risk premiums proportional to the level of volatility.

The paper reports the following empirical findings. First, we document that the simple strategy of just writing options (short) on the S&P 500 make an unusually

large Sharpe ratio. This evidence supports findings in Coval and Shumway (2001) who use data preceding ours. Second, the model based trading strategies work in that they provide higher Sharpe ratios than those obtained from just selling options. This is particularly so for options of short maturities. Third, returns from the options strategies have insignificant ‘betas’ with respect to returns on the underlying S&P 500 index, as well as the estimated spot volatility. If we interpret the market returns and the volatility innovations as systematic risk factors, we cannot reject the null hypothesis that the model based options returns have zero systematic risk.

This remainder of the paper is organized as follows. The next section discusses the construction of derivatives portfolios, discusses models to be considered, and discuss the computation of posterior expected utilities. Section 3 describes the data set. Section 4 presents empirical evidence on the performance of various trading strategies and discusses whether this is consistent with rational asset pricing. Section 5 concludes.

## 2 Optimal derivatives portfolios

The problem of choosing an optimal portfolio of derivatives is admittedly no different from the problem of choosing an optimal portfolio of “securities.” Traditionally, portfolio theory has been made in reference to stock markets, and the portfolio decisions are assumed to be based upon forecasts of mean return and return risk of those stocks in question. It is fair to say that the traditional portfolio theory has assumed that the investor possesses rather vague information about the future prospects of individual securities (stocks). Even if forecasting power is weak, portfolio diversification will enhance the signal to noise ratio in the forecasts. Hence, diversification may make up for noisy forecasts. Noisy forecasts are equivalent to uncertainty about the current value of the individual stocks.

The derivatives literature is characterized by an almost exactly opposite problem: While the prices of individual equity shares are at best very uncertain, the academic literature pertaining to the pricing of financial options on the very same equity shares, is typically very precise. If we take model assumptions as given, then derivatives price models produce an exact theoretical price. Whenever the model price differs from that of the market price, the interpretation of the traditional derivatives literature is that the difference is an arbitrage opportunity. Hence, an investor who is using the Black–Scholes model, and who has an infinitely strong belief in his estimate of market volatility, would be willing to bet his whole fortune on a deviation between the model price and the market price.

A pragmatic investor would off course never enter such a bet. There are at least three immediate reasons why: The arbitrage argument requires the investor to maintain a continuous riskless hedge portfolio. Trading costs, market closures, among other things, make this trading strategy infeasible in practice. And even if the riskless hedge existed, chances are great that the investor would be making a mistake relative to 1) the model itself, and 2) the parameter (volatility) for which an estimate is required.

In the following, we advocate a portfolio selection procedure that explicitly incorporates trading costs, discreteness of trading time, and parameter and model uncertainty.

## 2.1 Predictive distributions and Bayesian portfolio choice

A Bayesian decision maker is naturally concerned with the estimation inherent in model parameters. There is a large literature on Bayesian portfolio choice in financial markets. Early contributions focussed on finding analytic solutions analogous to standard analytical results in the classic Markowitz theory. In the classical approach, expected utility is computed conditional upon knowing the true parameter, so if  $w$  is the vector of portfolio weights, one seeks to maximize

$$E[U(w) | \Theta] = \int U(w)p(R | \Theta)dR,$$

where  $R$  is a return and  $\Theta$  is the model parameter. The Bayesian decision maker, by contrast, seeks to explicitly incorporate parameter uncertainty. Let  $p(\Theta | Y)$  denote a posterior distribution on the parameter  $\Theta$ . Then the Bayesian seeks to maximize

$$E[U(w) | Y] = \int \int U(w)p(R | \Theta)p(\Theta | Y)dRd\Theta.$$

The expectation is taken with respect to the *predictive* distribution,

$$p(R | Y) = \int p(R | \Theta)p(\Theta | Y).$$

More generally, in the case of derivative securities, we will also have to integrate over latent variables (such as volatility) to compute the predictive distribution.

In order to make the discussion precise, we discuss some models for options prices next.

## 2.2 The SVCJ model

The SVCJ model fall under the general class of affine jump diffusion models considered by [Duffie et al. \(2000\)](#). The acronym stands for “stochastic volatility with correlated jumps.” In this model, the stock price,  $S_t$ , is assumed to evolve according to the jump diffusion,

$$\frac{dS_t}{S_t} = \mu_S(t)dt + \sqrt{V_t}dW_t^S + dJ_t^S \quad (1)$$

$$dV_t = \kappa(\theta - V_t)dt + \sigma_V\sqrt{V_t}dW_t^V + dJ_t^V, \quad (2)$$

where jumps arrive with constant frequency  $\lambda$ , and jump sizes have joint distribution

$$Z_t^S | Z_t^V \sim N\left(\mu_y^Q + \rho_J Z_t^V, \sigma_S^2\right) \quad (3)$$

$$Z_t^V \sim \exp(\mu_V) \quad (4)$$

where  $V$  is the volatility process. The Brownian increments,  $dW^S$  and  $dW^V$  are correlated and  $E(dW_t^S dW_t^V) = \rho dt$ .

The jump term,  $dJ_t^i = Z_t^i dN_t^i$ ,  $i = \{S, V\}$ , has a jump-size component  $Z_t$  and a component given by a Poisson counting process  $N_t$ . In the case of a common Poisson process,  $N_t^V = N_t^S$ , jump sizes  $Z_t^S$  and  $Z_t^V$  can be correlated. The specification of jump components determines the models to be considered. We discuss the two possible specifications next.

The basic stochastic volatility (SV) model with a square root diffusion driving volatility, was initially proposed for option pricing purposes in [Heston \(1993\)](#). It obtains as a special case of the general model in this paper with jumps restricted to zero ( $dJ_t^S = dJ_t^V = 0$ ). The volatility process,  $V_t$ , captures serial correlation in volatility. The  $\kappa$  and  $\theta$  parameters measure the speed of mean reversion and the (mean) level of volatility, respectively. The parameter  $\sigma_V$  is commonly referred to as the “volatility of volatility”. Higher values of  $\sigma_V$  imply that the stock price distribution will have fatter tails. The correlation parameter,  $\rho$ , is typically found to be negative which implies that a fall in prices usually will be accompanied by an increase in volatility which is sometimes referred to as the “leverage effect” ([Black 1976](#)). A negative  $\rho$  implies that the conditional (on initial stock price,  $S_t$ , and volatility,  $V_t$ ) returns distribution is skewed to the left.

### 2.3 Computing predictive densities

In the following, we describe a simple numerical scheme to estimate the predictive densities, or equivalently, a scheme that allows computation expectations of (utility) functions with respect to the predictive distribution.

Let  $\Theta^{(g)}$ ,  $V_t^{(g)}$  denote posterior samples of the parameter vector and the latent volatility. It is assumed that such samples are available. A number of schemes involving Gibbs and Metropolis Hastings, possibly in conjunction, are available for this purpose.

Assuming that the posterior density is formed based information,  $Y_t$  as of time  $t$ , we wish to compute  $E[U(w) | Y_t]$ . The sample mean  $\hat{E}[U(w) | Y_t] = \frac{1}{G} \sum_g U(w)^g$ , where  $U(w)^g$  denotes the utility given the  $g$ th posterior draw  $\Theta^{(g)}$ ,  $V_t^{(g)}$ , is a simulation consistent estimator under weak regularity.

### 2.4 Filtering

A final computational, statistical problem remains to be considered. As time progresses and market conditions change, we will typically like to re-balance the portfolio. Naturally, for any time  $t$  we can form a new posterior, and the procedure outlined above can be repeated. The computation of posterior densities by simulation is typically very time consuming and it can be useful to consider keeping the posterior for the parameter fixed, while updating the predictive density for the latent state variable,  $V_t$ .

This calls for solving a filtering problem, which is tantamount to designing an MCMC sampler that draws

$$V_t \sim p(y_t | V_t, \Theta)p(V_t | V_{t-1}, \Theta),$$

for  $t = 1, \dots, T$  where  $y_t$  denotes the  $t$  realization of derivatives data.

Notice that for parameter estimation, we draw  $V_t, t = 1, \dots, T$  from the posterior

$$p(y_t | V_t, \Theta)p(V_t | V_{t-1}, \Theta)p(V_{t+1} | V_t, \Theta)$$

Hence: We can solve the filtering problem by just removing one term from the expression used for the posterior simulation. Sampling from this density is a straightforward operation<sup>1</sup> and can be carried out using standard sampling schemes such as Metropolis Hastings.

## 2.5 Optimal portfolios with fixed planning horizon

Having outlined how to deal with the posterior averaging, we now describe how to make portfolio decisions in the presence of trading costs. We are only considering trading costs in the form of bid/ask spreads. Other types of costs, such as brokerage fees, typically only apply at the retail level. Investment firms with reasonably large holdings face negligible additional costs.

This paper is concerned with investment decisions of an investor who has a fixed planning horizon. This is a simplification of the dynamic programming problem which leads to choices over portfolios and expected holding times.

Let  $n_t^{\text{long}}$  and  $n_t^{\text{short}}$  denote vectors of portfolio holdings with the  $i$ th element representing the number of shares held of security  $i$ . The value of a portfolio is reasonably measured as the value obtained if positions were unloaded at prevailing market prices through market orders. So portfolio value is measured at long/bid and short/ask

$$W_t = n_t^{\text{long}} S_t^{\text{bid}} + n_t^{\text{short}} S_t^{\text{ask}} + K_t \quad (5)$$

where  $K_t$  is the cash holding. We can write the change in portfolio value,  $\Delta V_{t+1} = V_{t+1} - V_t$ , as

$$\Delta W_{t+1} = n_t^{\text{long}} \Delta S_{t+1}^{\text{bid}} + n_t^{\text{short}} \Delta S_{t+1}^{\text{ask}} + K_t r \quad (6)$$

where  $r$  represents a constant per period rate of interest.

Define changes in portfolio holdings (i.e., trade) to be  $\Delta n_t$ , and decompose it into shares bought  $\Delta n_t^+ = \Delta n_t I(\Delta n_t > 0)$  and shares sold  $\Delta n_t^- = \Delta n_t I(\Delta n_t < 0)$ . The product of the  $\Delta n_t$  with the indicator  $I(\cdot)$  is element by element.

<sup>1</sup> The sampler outlined here solves essentially the same filtering problem as the algorithm known as the “particle filter.” Particle filtering nevertheless suffer from the unfortunate property that they will eventually reject all but one of the sampled values  $\Theta^{(g)}$  as  $t$  increases. The particle sampler, therefore, does not converge.

The dynamics of the relevant state variables are

$$W_{t+1} = n_{t+1}^{\text{long}} S_{t+1}^{\text{bid}} + n_{t+1}^{\text{short}} S_{t+1}^{\text{ask}} + K_{t+1}, \quad (7)$$

$$K_{t+1} = (K_t - \Delta K)(1 + r), \quad (8)$$

$$n_{t+1} = n_t + \Delta n_t, \quad (9)$$

$$\Delta K = \Delta n_t^+ S_t^{\text{ask}} + \Delta n_t^- S_t^{\text{bid}}. \quad (10)$$

Trading decisions are found by maximizing the expected utility

$$\Delta n_t = \operatorname{argmax}_t E_t [U(W_{t+\tau})] \quad \text{s.t. } \Delta n_t \in \mathcal{S}$$

where the expectation is taken with respect to the predictive density for  $S_{t+\tau}$ , and  $\mathcal{S}$  is a set of tradable securities. For example, in the empirical study below we allow trading only in securities for which there are valid bids and asks. Conditioning upon the possibility of missing values in this way is necessary because options contracts have a tendency to stop trading as the underlying price changes in such a way that the contract becomes far in, or far out of the money. This is an overriding concern. There are other practical limitations, which also require special considerations. One prominent problem is finding a unique solution to the portfolio problem. This may be difficult if the covariance matrix of option payoffs is singular, or near singular. For example, theoretical prices of puts and calls with the same maturity and strike are perfectly collinear, giving a singularity. Two ways of dealing with this are, 1) each option is assumed measured with idiosyncratic noise, and 2) puts and calls are traded only as singletons or in pairs, as in straddle positions. The empirical study is accordingly conducted using out-of-the-money puts and calls, as well as at-the-money straddles. Here we impose the straddle restriction if the “quasi  $t$  ratio”  $t = \sqrt{\tau/V_0}(X/S - 1)$  falls within a specified range  $R$ . Here the factor  $\tau/V_0$  is the  $\tau$  period variance assuming that the volatility remains at its initial value,  $V_0$ , and  $X/S - 1$  is the moneyness.

This study assumes that options must be bought and sold at market posted asks and bids respectively. The bid–ask spreads in options markets are rather large, which means that our estimates of trading profits will be very conservative. This is so because investors typically can transact inside the bid/ask spread by posting their own bids and offers. This is true of, for example, market makers. To see how decisions differ for an investor who can transact at the mid-point of the bid/ask, we simplify the system above assuming  $S_t^{\text{bid}} = S_t^{\text{ask}} = S_t$ , which gives

$$W_{t+1} = (n_t + \Delta n_t) S_{t+1} + (K_t - \Delta n_t S_t)(1 + r). \quad (11)$$

Returns are given as,

$$\frac{W_{t+1}}{W_t} = \frac{(n_t + \Delta n_t) S_{t+1} + (K_t - \Delta n_t S_t)(1 + r)}{n_t S_t + K_t}. \quad (12)$$



In this case, an investor with mean variance preferences  $U(R) = E(R) - \frac{1}{2}\gamma \text{Var}(R)$  over returns,  $R$ , will optimally revise his portfolio according to the rule,

$$\Delta n_t = \frac{1}{\gamma} \text{Cov}(S_{t+1})^{-1} (E_t S_{t+1} - S_t(1+r)) - n_t.$$

The assumption of zero transactions costs is not justified in the empirical study considered here, as the transactions costs in the form of bid/ask spreads are significant.

### 2.5.1 Testing of Sharpe ratios

The empirical study below exclusively focuses on Sharpe portfolio measures. Consistent with the Bayesian view of data analysis, one might suggest constructing posterior distributions on the Sharpe ratios. This is not possible. The out-of-sample performance study is done while explicitly averaging over estimated parameters and latent variables. Indeed, the Sharpe ratios are formed by constructing returns data from the predictive density  $p(R_t | R_{[0:t-1]})$  which does not depend upon the parameters. Since it is impossible to construct posterior distributions on these performance measures, we appeal to standard sampling theory and construct frequentist standard errors on the Sharpe ratios<sup>2</sup>.

A test for the null hypothesis of equal Sharpe ratios between two portfolios is easily carried out. In the following, I describe a test statistic which is a slight generalization of the test in [Mommel \(2003\)](#). This generalization allows the first two moments of the sampling distribution to be estimated based on bootstrapping which makes the test robust to autocorrelation and heteroscedasticity.

The null hypothesis is  $s_1 = s_2$  for  $s_i = (\hat{E}(r_i) - rf)/\hat{\sigma}_i$  so the distribution of  $z := s_1 - s_2$  forms the basis of a standard test. Notice that if the two portfolios have the same standard deviation, then the test statistic is positive if and only if  $\hat{E}(r_1)$  is significantly greater than  $\hat{E}(r_2)$  which can be tested using a standard  $t$  test. For two return series with different standard deviations, assume that an investor take has weights  $w_1 = 1$  and  $w_2 = \sigma_1/\sigma_2$  in the two assets. Then the portfolio returns have standard deviations equal to  $\sigma_1$ . If the return differential  $d = r_1 - r_2\sigma_1/\sigma_2$  is significantly positive, we conclude that  $s_1$  is significantly greater than  $s_2$ , and vice versa. Since the test now reduces to the problem of testing for the mean location of  $d$ , we can use a variety of methods robust to heteroscedasticity and autocorrelation. Here, we apply a circular block bootstrap to estimate the sampling variability of  $\hat{E}(d)$ .

## 3 Data

This paper uses data from “optionmetrics” made available through the Wharton Research Data Services, <http://wrds.wharton.upenn.edu>. The data set consists of closing bid and asking prices for S&P 500 cash options collected daily between January 4, 1996, and February 28, 2003. There are a total of 167,551 put options and 117,303 call

<sup>2</sup> This point was also noted by [Johannes et al. \(2002\)](#) in the context of dynamic trading strategies for stocks.

**Table 1** Data summary

|                      | Puts  | Calls |
|----------------------|-------|-------|
| Mid bid–ask price    | 31.23 | 49.37 |
| Intrinsic value      | 12.70 | 24.08 |
| Moneyiness ( $X/S$ ) | 0.925 | 1.031 |
| Maturity             | 89    | 79    |
| Bid–ask spread       | 1.32  | 1.45  |

**Table 2** Descriptive statistics for option returns

|          | Puts     | Calls   |
|----------|----------|---------|
| Mean     | −0.00822 | 0.00879 |
| SD       | 0.320    | 0.396   |
| Min      | −0.921   | −0.958  |
| Max      | 8.66     | 15.95   |
| Skewness | 4.41     | 7.54    |
| Kurtosis | 65.6     | 181.1   |

options for which there are simultaneously quoted bids and asks. In the analysis which follows, only options with both bids and asks posted are used. Table 1 presents some key numbers. The mean put price is 31.2 with an intrinsic value of 12.7, giving an average premium of 18.3. The mean bid–ask spread is 1.32, some 7 % of the average premium. For the calls, the bid–ask spread is about 5.6 % of the premium.

If we assume that the fair value of the option is the mid-point of the bid and the ask, a trader who holds a position until expiration would have to expect in excess of a 3.5 % return on a put just to break even. A trader who plans to sell a put before expiration would have to have an expected return exceeding the 7 % spread to break even.

These are obviously large numbers. They indicate that any investor who is issuing market orders is at a large disadvantage relative those taking the opposite sides of these bets. The Sharpe ratios computed below, therefore, should be interpreted as lower bounds on attainable Sharpe ratios for investors who can transact inside the bid/ask bounds, either because they issue their own limit orders, or because they are market makers.

## 4 Empirical results

We start the empirical exploration by re-examining evidence of consistent high returns to options writers. Table 2 reports descriptive statistics for the options data. The data on display are changes in the midpoint of the bid and the ask of the data. Price changes are measured as a percentage of the previous days price (i.e., an arithmetic return). The data are recorded whenever there is both a bid and an ask posted. Multi-day returns are recorded whenever consecutive bid or ask quotes are missing.

As evidenced in the table, put options changed by an average of −0.822 % per period (typically 1 day). The corresponding number for calls is 0.879 %. It is hard to interpret these numbers because they do not in anyway extrapolate into long term

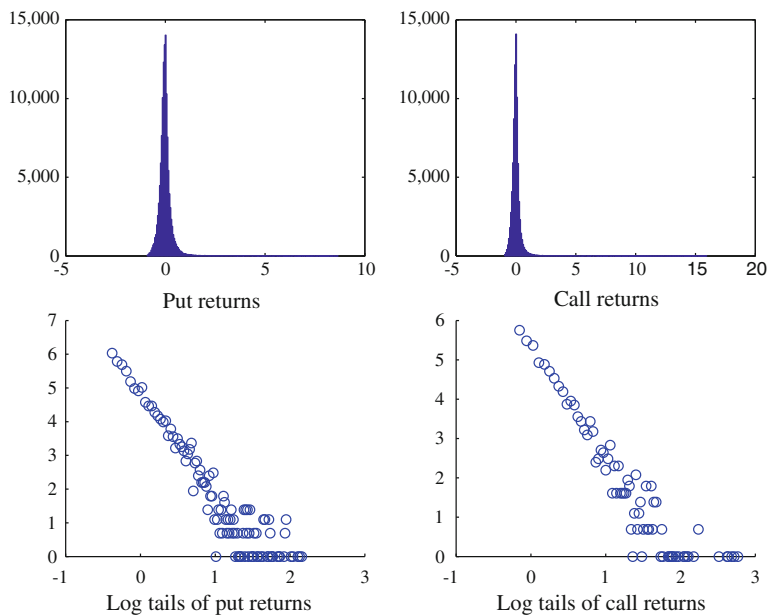
performance of option writers/buyers. For example, the standard practice of approximating the annual return by multiplying by 252 trading days in a year would seem to indicate that both a call buyer and a put writer would earn a staggering 200 % annually. This is a standard way of interpolating from daily into multi-day returns. For example, Coval and Shumway report that “An at-the-money call option tends to earn an average return between 1.85 and 2.00 % per week, or between 96 and 110 % on a annualized basis.” Hence, the annualized average returns reported here are on a level with Coval and Shumway’s estimates. They, in fact, also report daily returns which are close to those of Table 2.

#### 4.1 The problem with arithmetic returns

Unfortunately, the 200 % annual returns are not real. The problem is the sum of two one-day returns does not equal the two day return. For example, a 40 % increase followed by a 40 % decrease in an assets value gives an average return of 0, but leaves the investor with a 16 % loss. Notice that a 40 % daily percentage change in the value of a call is exactly what an investor should expect, as the daily standard deviation is 0.396. If an asset realizes consecutive returns of 1 and  $-1$  %, the actual two period return is about  $-0.01$  %, and thus close to the arithmetic average of zero. What these numbers illustrate is that the usual practice of multiplying a daily return by the number of trading days (252) to construct an annualized return, leads to an extremely upwardly biased annualized estimates for assets with high volatility. While this is true for regular stocks with high volatilities, the situation is extreme for stock options for which the daily volatility is in excess of 20 times the underlying index volatility, as well as extremely heavy tailed. Thus we conclude, the mean daily returns in Table 2 do not necessarily predict high annualized returns to options traders. This does not mean that writing options is not profitable, it only means that the daily returns are poor measures of long term performance.

#### 4.2 Do moments of options return exist?

Figure 1 reveals another problem with return distributions—an extreme right tail. For example, of the 124 thousand observations, there are 478 put returns larger than 159 %—a five standard deviation event. The maximum put and call returns are 866 and 1,595 % respectively. These are 27 and 40 standard deviation events. With tails this heavy, it is an open question whether any of the moments of the return distributions actually exist. To shed some light on this, assume that the tail of the return density follows a power law;  $p(R_c) \propto ax^b$ . If  $b$  is greater than  $-2$ , the expectation does not exist since  $\int^\infty x \times ax^{-2}dx$  diverges. Similarly,  $b$  needs to be lower than  $-3$  for the variance to exist. Figure 1 show the log-tails of the estimated frequency histograms. Linear regression estimates of  $b$  are  $-2.59$  (0.095) and  $-2.22$  (0.1) which produces  $t$  statistics equal to 6.2 and 2.12 for the null hypothesis that the  $b$ ’s are greater than  $-2$ . Thus, there is little evidence against the hypothesis that the means exist. We can similarly reject a null hypothesis that the unconditional variance, as well as higher order moments, exist. This also spells trouble for standard methods of inference about



**Fig. 1** Density *histograms* and estimated log-tails for daily options returns

the mean returns. For example,  $t$  statistics for testing the location of the mean are unreliable because they rely on the gaussian central limit theorem whose assumptions are violated (variance need to exist).

#### 4.3 Performance of option writers

Coval and Shumway also report monthly returns to a trading strategy consisting of selling straddles at market prevailing bids. To offset the effects of occasional large drops, they augment the simple sell strategy with an offsetting position in an out-of-the-money put, bought at market offered prices. The strategies yield a monthly return of about 3–4 %, with a Sharpe ratio of about 0.3. The corresponding annualized numbers are 12–16 %, and 1. These are on the order of twice the historically attainable Sharpe ratio from a buy and hold position in the underlying S&P 500 index.

Table 3 presents evidence of the annual Sharpe ratio earned by writing options in over the 1996–2003 period. The Sharpe ratio earned over the whole sample is 0.457. This compares to about 0.2 for the S&P 500 over the same period. Thus, we conclude that writing options, although not as attractive as reported for the 1988–1996 period, still beats buying the S&P 500. The table also reveals that the reduction in the attainable Sharpe ratio from writing options over the whole sample is likely due to relatively cheap options markets over the 1996–1999 period. The subsequent 1999–2003 period yields a Sharpe ratio of 0.86.

Figure 2 depicts the performance of options writers through four key time series. The bottom plot shows the portfolio value of an options writer. As can be seen, the

**Table 3** Performance of options writers

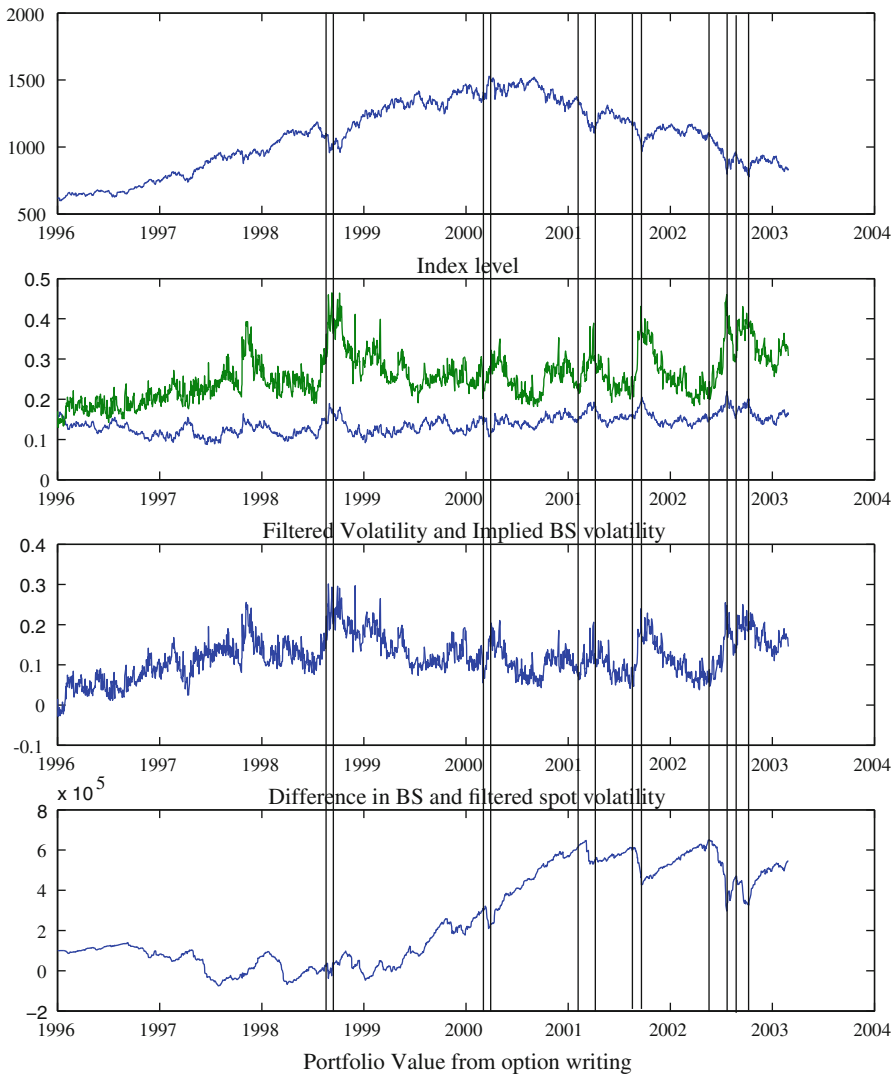
|           | Puts    | Calls   | All     |
|-----------|---------|---------|---------|
| 1996–2003 |         |         |         |
| Sharpe    | 0.238   | 0.077   | 0.457   |
| SE        | (0.058) | (0.056) | (0.055) |
| 1996–1999 |         |         |         |
| Sharpe    | 1.325   | −1.339  | −0.374  |
| SE        | (0.124) | (0.123) | (0.120) |
| 1999–2003 |         |         |         |
| Sharpe    | −0.196  | 0.990   | 0.860   |
| SE        | (0.095) | (0.093) | (0.090) |

The table reports Sharpe ratios for portfolios of short options positions. Trading is done by selling at best bids. Bootstrapped standard errors in parenthesis

options writer makes a positive profit on average; however, there is a long period of dismal returns characterizing the '96–'99 period. The Sharpe ratio earned by writing options over this period is  $-0.37$ , as shown in Table 3. The losses can perhaps naively be characterized as losses on call options which ended in-the-money during this period with a substantial increase in the value of the S&P 500. Another indicator of option valuation levels is given by the Black–Scholes implied volatilities. The second plot in Fig. 2 shows implied volatility of short dated options as well as spot volatility obtained by the filtering algorithm described above. The third plot shows the difference between these two. Casual observation from these graphs reveal a few interesting patterns. First, implied volatility is consistently higher than spot volatility. Second, during periods in which the difference between the volatilities (henceforth “volatility spread”) increases, options writers tend to lose money. For example, over the period from January 1996 to about September 1998 (first vertical line in the graph), options writers are losing money. Over the same period the spread is increasing from negative to about 0.25. These numbers are annualized, and correspond to a difference of opinion of about 1.5 % daily standard deviation. Put in perspective, this roughly means that the option market seems to believe that the daily standard deviation is about 2.5 %, while filtered spot volatility is close to the historical average for the S&P 500 of about 1 % daily.

Obviously, these deviations are large and may suggest that the market at the time was overvaluing the options. The following period from Fall of 1998 to Spring 2001 proved very profitable for options writers, as evidenced by the bottom plot in Fig. 2. If we accept that the volatility spread is a rough measure of option overpricing, the size of the spread at the beginning of the period, as well as the fact that the spread shrunk from 0.25 to about 0.1, are two possible explanations for the large returns over this period (the Sharpe ratio is about 1.5).

The vertical lines in the plot mark periods in which the option portfolio is losing relatively much over a short period. During most of these periods, it is possible to make a visual link between the reduction in option portfolio values, and changes in either the volatility spread, or the stock price. For example, during the spring and fall of 2001, both the S&P index and the volatility spread dropped (increased) sharply, leading to sharp declines in the portfolio of short options.



**Fig. 2** Time series of index levels, returns, volatility, and profits from writing options

#### 4.4 Model estimation

In order to compute theoretical options prices we need estimates of the parameters that govern the dynamics of the underlying asset. The literature on estimation for jump diffusions is sizable and includes (Gourieroux et al. 1993; Gallant and Tauchen 1996; Andersen et al. 2002), among others. Bakshi et al. (1997), Bates (1996, 2000), and Pan (2002) consider estimation of model parameters from options prices. Estimation of the SVCJ model and the SV model has been discussed at length in Eraker et al. (2003), and in Eraker (2004). Table 4 reproduces results from Eraker (2004) on summary statistics of MCMC samples from a posterior density formed based on daily returns data

**Table 4** Parameter estimates

|            | SV                                    | SVYJ                                  | SVCJ                                  |
|------------|---------------------------------------|---------------------------------------|---------------------------------------|
| $\alpha$   | 0.026<br>(0.011)<br>[0.000, 0.051]    | 0.026<br>(0.011)<br>[0.000, 0.050]    | 0.030<br>(0.011)<br>[0.004, 0.056]    |
| $\theta$   | 0.881<br>(0.098)<br>[0.692, 1.163]    | 0.834<br>(0.122)<br>[0.590, 1.221]    | 0.573<br>(0.078)<br>[0.397, 0.750]    |
| $\kappa$   | 0.017<br>(0.005)<br>[0.008, 0.030]    | 0.012<br>(0.006)<br>[0.004, 0.022]    | 0.016<br>(0.003)<br>[0.009, 0.023]    |
| $\rho$     | -0.373<br>(0.056)<br>[-0.500, -0.242] | -0.468<br>(0.065)<br>[-0.601, -0.295] | -0.461<br>(0.073)<br>[-0.616, -0.237] |
| $\sigma_V$ | 0.108<br>(0.011)<br>[0.082, 0.137]    | 0.079<br>(0.011)<br>[0.061, 0.104]    | 0.058<br>(0.012)<br>[0.030, 0.078]    |
| $\mu_y$    |                                       | -3.661<br>(2.486)<br>[-10.752, 1.281] | -3.225<br>(2.523)<br>[-10.086, 2.436] |
| $\rho_J$   |                                       |                                       | 0.312<br>(1.459)<br>[-3.580, 3.833]   |
| $\sigma_y$ |                                       | 6.628<br>(1.697)<br>[3.714, 11.742]   | 4.918<br>(1.272)<br>[2.880, 9.295]    |
| $\mu_V$    |                                       |                                       | 1.250<br>(0.381)<br>[0.681, 2.523]    |
| $\lambda$  |                                       | 0.003<br>(0.001)<br>[0.001, 0.006]    | 0.004<br>(0.001)<br>[0.001, 0.007]    |

The table reports posterior means and standard deviations (in parenthesis), and 99 % credibility intervals (in square brackets) for parameters in the jump diffusion models based on returns data only. Parameter estimates were obtained using the estimation procedure in [Eraker et al. \(2003\)](#) using 5,307 time-series observations of the S&P 500 index from January 1970 to December 1990

from 1970 to 1990. The results in Table 4 have previously been reported in [Eraker \(2004\)](#). They were obtained prior to the current study, and the results which follow are therefore true out-of-sample results. The results in Table 4 have been thoroughly discussed elsewhere, and readers are referred to the above mentioned papers for an interpretation of the estimates.

#### 4.5 Performance of model based investment and hedging strategies

Table 5 tabulates Sharpe ratios from dynamic portfolio allocations based on the deviations between market and model prices. The table presents the performance of a “simple” strategy which calls for buying/selling one unit of under/overpriced assets, vs. the hedging strategies. The strategies are formed for specific maturity categories.

**Table 5** Model based investment and hedging performance

| Maturity category | SV               |                  | SVCJ             |                  |
|-------------------|------------------|------------------|------------------|------------------|
|                   | No hedge         | Hedge            | No hedge         | Hedge            |
| 1                 | 0.298<br>(0.056) | 1.111<br>(0.056) | 1.002<br>(0.056) | 1.187<br>(0.054) |
| 2                 | 0.963<br>(0.055) | 1.289<br>(0.055) | 1.221<br>(0.055) | 1.438<br>(0.054) |
| 3                 | 1.200<br>(0.054) | 1.367<br>(0.056) | 1.546<br>(0.056) | 1.603<br>(0.056) |
| 4                 | 0.761<br>(0.056) | 0.858<br>(0.054) | 0.900<br>(0.055) | 0.971<br>(0.056) |
| 5                 | 0.298<br>(0.055) | 0.464<br>(0.055) | 0.524<br>(0.056) | 0.508<br>(0.057) |
| All               | 0.735<br>(0.053) | 1.140<br>(0.057) | 1.192<br>(0.056) | 1.323<br>(0.057) |

The table reports realized Sharpe ratios for portfolios of options formed conditional upon either the stochastic volatility model (SV), or the stochastic volatility with correlated jumps (SVCJ) model. Trading is done through market orders, buying at the asking prices and selling at bids. Bootstrapped standard errors in parenthesis

The first category consists of the shortest maturities over 12 days, the second is the second shortest, and so on.

As can be seen from the table, the Sharpe ratios are relatively high, ranging from 0.29 to 1.6 for the respective maturity/model combinations. By contrast, the historical Sharpe ratio of the S&P 500 index is about 1/2. The Sharpe ratios are generally greater for the SVCJ model, and generally greater when the hedging strategy is employed.

The statistical significance<sup>3</sup> hedging and model choice is evaluated in Tables 6 and 7. The tables contain pairwise significance tests for the null hypothesis that the Sharpe ratios of two portfolio strategies are identical. For example, Table 6 reports a  $z$  statistic of 1.688 which indicates that the performance of the hedged portfolio with short maturity options based on the SV model is significantly better than the un-hedged and 1.111 is significantly better than 0.298. As can be seen from the table, the difference between Sharpe ratios of un-hedged and hedged portfolios in Table 5 are smaller. This produces less significant test statistics in Table 6. However, for the SV model, the hedging strategy is found overall to produce significantly better results. This is not true for the SVCJ model, for which the overall  $z$ -statistic is 0.886. The improvement in performance of hedging for the SV model and not the SVCJ model may be related to the fact that the SV model inherently generates less tail mass at shorter maturities

<sup>3</sup> The pairwise comparisons tests are based on asymptotic normality of sample means. Specifically, if  $r_1$  and  $r_2$  are two return series, and  $z = (r_1/s_1 - r_2/s_2)/\sqrt{s_1^2 + s_2^2}$  where  $s_i = \text{std}(r_i)$  is the sample standard deviation of  $r_i$ , then the test of the null is constructed by testing whether  $\sum z_t = 0$ . This test is conducted using a circular block bootstrap which accounts for non-normality and time varying volatility.



**Table 6** Significance of hedging

| Maturity category | SV      | SVCJ   |
|-------------------|---------|--------|
| 1                 | 1.688** | 1.016  |
| 2                 | 1.600*  | 1.102  |
| 3                 | 0.748   | 0.306  |
| 4                 | 0.456   | 0.394  |
| 5                 | 0.970   | −0.172 |
| All               | 1.818** | 0.886  |

The table reports pairwise tests of the null hypothesis that there is no difference in the Sharpe ratios of hedged and un-hedged portfolios

\* denote significance at the 10 % level

\*\* denotes significance at the 5 % level

**Table 7** Significance of model choice

| Maturity category | No hedge | Hedge    |
|-------------------|----------|----------|
| 1                 | 0.655    | 1.492*   |
| 2                 | 1.375*   | 1.850**  |
| 3                 | 2.344*** | 2.158*** |
| 4                 | 1.391*   | 1.694**  |
| 5                 | 0.391    | 1.273    |
| All               | 2.269*** | 3.007*** |

The table reports pairwise tests of the null hypothesis that there is no difference in portfolio choice made using the SV and SVCJ models

\* denotes significance at the 10 % level

\*\* denotes significance at the 5 % level

\*\*\* denotes significance at the 1 % level

than does the SVCJ model. Bayesian parameter averaging offsets this, as the predictive distributions. This may explain the improvement for the SV model.

Table 7 compares the SVCJ model with the SV model. In short, the SVCJ model is found to outperform the SV model, and particularly so for the hedged portfolios. This is perhaps not so surprising as the SVCJ model has been shown to provide a significantly better description of the underlying S&P 500 returns (see [Eraker et al. 2003](#)).

#### 4.6 Can volatility and jump risk premia explain high returns?

The issue of essential interest in what follows is whether the high rates of return documented in the preceding section can be explained by rational risk premia related to systematic factors. In addressing this issue, we proceed in two distinct ways. Along the lines of classical asset pricing theory tests, the paper examines the results of regressions of portfolio returns onto systematic risk proxies—in this case the equity returns and innovations in the volatility factor.

**Table 8** Trading return sensitivity to S&P 500 returns and volatility

|               | Const            | S&P return        | Volatility changes |
|---------------|------------------|-------------------|--------------------|
| A: No hedging |                  |                   |                    |
| SV            | 0.009<br>(0.005) | 0.424<br>(0.390)  | −0.003<br>(0.028)  |
| SVCJ          | 0.044<br>(0.022) | −1.572<br>(1.793) | 0.000<br>(0.127)   |
| B: Hedging    |                  |                   |                    |
| SV            | 0.099<br>(0.044) | −3.563<br>(3.583) | −0.035<br>(0.254)  |
| SVCJ          | 0.088<br>(0.029) | −2.225<br>(2.363) | −0.012<br>(0.168)  |

The table reports posterior means and standard deviations for the regression of returns from trading strategies on S&P returns and volatility changes. Volatility factor loading incorporate first step estimation uncertainty

Evidence presenting the sensitivity of trading returns to S&P 500 returns and volatility changes is presented in Table 8. These regressions should be interpreted in a Bayesian way, and the estimates represent posterior means of linear regression coefficients, averaged against the filtering distribution of spot volatility.

As can be seen from the table, there is basically zero correlation between the portfolio returns, and the candidate risk factors. All the factor loadings are statistically insignificantly different from zero. On the other hand, 95 % posterior credibility intervals or frequentist confidence intervals assign significant probability to non-trivial risk premiums. In essence, the results in Table 8 are inconclusive as to the presence of a volatility risk premium. From the table, we see that the market “beta’s” are estimated to be negative for three out of four portfolios. While insignificant, this result in isolation seems to indicate that if anything, the high rates of return on the option portfolios is not due to positive market risk.

It is important to realize that the trading strategies examined in this paper are formed using model prices *which assume a zero risk premium for bearing volatility and jump risk*. This was done on purpose, and affects the trading strategies such that on average, the expected returns to selling straddles are greater than if we computed model prices using positive premiums. This is because the theoretical option prices are positively dependent upon these risk premium parameters. Thus, by increasing these parameters, it is possible that the theoretical prices approach, on average, the market prices. If so, the trading strategies will on average be neither long nor short, consequently leading to random returns which should average zero absent trading costs and be negative considering trading costs.

The SVCJ model contains two types of risk premia and we will discuss both in turn. Within the framework of affine jump diffusion models, standard risk premiums are associated with the diffusive noise terms. A market risk premium is attached to  $W^S$  the Brownian motion driving the stock returns, and a volatility risk premium is associated with the Brownian motion driving volatility,  $W^V$ . The market price of volatility risk is proportional to volatility  $\sqrt{V_t}$  such that the volatility drift is

**Table 9** Effects of volatility risk premium

| Maturity category | 1     | 2     | 3     | 4     | 5     | All   |
|-------------------|-------|-------|-------|-------|-------|-------|
| Sharpe ratio      | -0.43 | -0.44 | -0.56 | -0.57 | -0.28 | -0.42 |
| Average holdings  | 0.73  | 1.32  | 3.49  | 10.53 | 23.48 | 39.56 |

The table reports realized Sharpe ratios for portfolios of options formed conditional upon stochastic volatility with correlated jumps (SVCJ) model using a volatility risk premium of 0.011

$$\mu_V(V_t) = \kappa^Q(\theta^Q - V_t), \quad (13)$$

$$\kappa^Q = \kappa - \eta_V \quad (14)$$

$$\theta^Q = \frac{\kappa\theta}{\kappa - \eta_V} \quad (15)$$

under the equivalent Martingale measure. The parameter  $\theta^Q$  is the long term, average volatility under the risk neutral measure. As easily seen, the average  $Q$  volatility explodes as  $\eta_V$  approaches  $\kappa$  from below. This means, in effect, that theoretical option prices can be made arbitrarily expensive by increasing the  $\eta_V$  to a value close to  $\kappa$ . Since theoretical prices depend positively upon  $\theta^Q$ , we might expect that there exists a numerical value of  $\eta_V$  such that

- The average theoretical value of options contracts can be made arbitrarily large
- Portfolio decisions based on the discrepancy between market and model prices can be reversed by reversing the sign of the pricing error
- Portfolio profits from model based strategies approach zero, or go negative

Table 9 tabulates the Sharpe ratios earned by the portfolio strategies where the theoretical model prices have been computed assuming a volatility risk premium of  $\eta_V = 0.011$ . This value was estimated in Eraker (2004). Apart from the impact of  $\eta_V$  on the  $\kappa^Q$  and  $\theta^Q$ , the numbers were computed as in the rightmost column of Table 5. The numbers in Table 9 clearly illustrate the empirical impact of the risk adjustment: trading profits turn negative (as in negative sharpe ratios), which is attributed to average positive holdings across all maturity categories. Thus, we conclude, increasing the parameter  $\eta_V$  from zero to 0.011 has, at least at a mechanical level, the effect of eroding the trading profits.

Seemingly our results above should be interpreted as support for a risk based explanation to the high returns/risk relationships offered by options investments. However, the conclusion still warrants an investigation into what may or may not comprise a reasonable reward to bearing the risks associated with changes in market volatility. To better gauge the effects of the volatility risk premium,  $\eta_V$ , on expected options returns, we present two additional pieces of evidence.

First, Table 10 tabulates prices of straddles for different values of  $\eta_V$ . The effects of varying this parameter is striking. At for example  $\eta_V = 0.011$ , the price of a 20 day option increases from 3.8 cents to 4.7—an almost 24 % increase. Similarly, 60 day options are almost 40 % more expensive with  $\eta_V = 0.011$ . These are obviously sizeable price-impacts for these short maturity contracts and may be expected to yield significant improvements in reward to risk ratios.

**Table 10** Impact of  $\eta_V$  on option prices

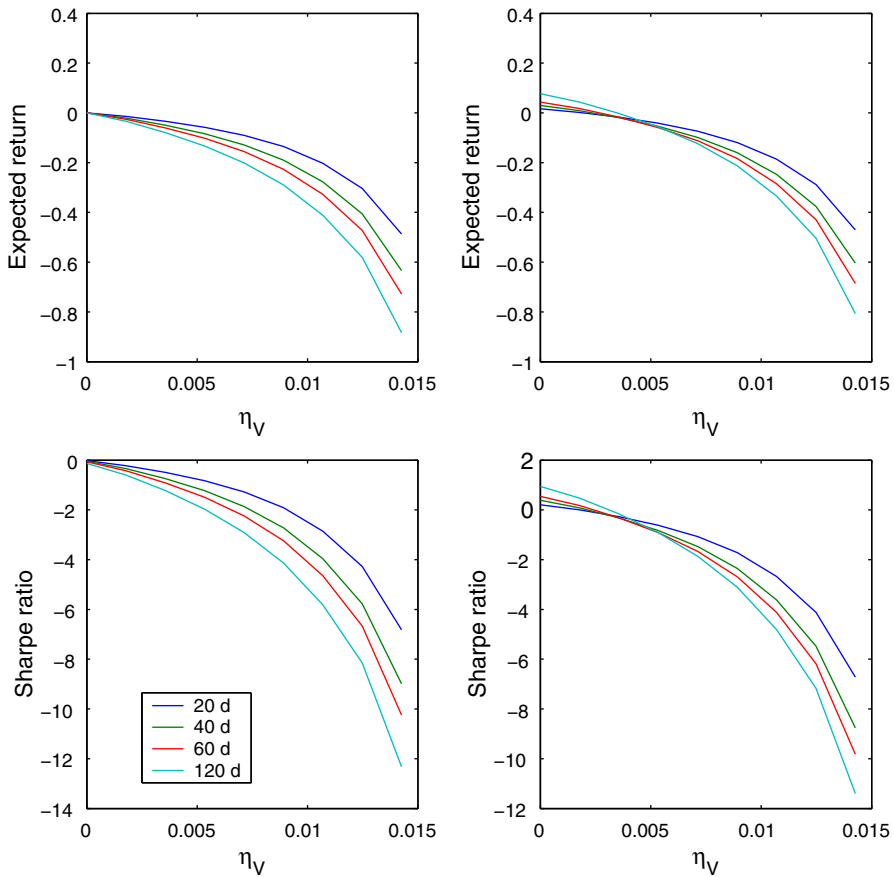
| $\eta_V$ | Maturity (days) |       |       |       |       |       |
|----------|-----------------|-------|-------|-------|-------|-------|
|          | 20              | 40    | 60    | 120   | 250   | 500   |
| 0.000    | 0.038           | 0.054 | 0.066 | 0.092 | 0.132 | 0.186 |
| 0.002    | 0.039           | 0.055 | 0.068 | 0.096 | 0.137 | 0.194 |
| 0.004    | 0.039           | 0.057 | 0.070 | 0.100 | 0.145 | 0.205 |
| 0.005    | 0.040           | 0.059 | 0.073 | 0.106 | 0.154 | 0.218 |
| 0.007    | 0.042           | 0.061 | 0.077 | 0.113 | 0.167 | 0.236 |
| 0.009    | 0.044           | 0.065 | 0.083 | 0.123 | 0.184 | 0.261 |
| 0.011    | 0.047           | 0.071 | 0.091 | 0.139 | 0.211 | 0.297 |
| 0.012    | 0.052           | 0.081 | 0.106 | 0.165 | 0.253 | 0.352 |
| 0.014    | 0.062           | 0.102 | 0.136 | 0.224 | 0.351 | 0.466 |

The table reports the theoretical values of and ATM straddles as a function of the volatility risk premium,  $\eta_V$ . Parameters are as in Table 4, interest rate is 2%, initial stock price is \$1

Quantifying the effects of  $\eta_V$  further, we present in Fig. 3 expected rates of return and Ex ante Sharpe (reward-to-variability) ratios for straddle buyers as a function of  $\eta_V$ . The figure depicts the expected rates of return and Sharpe ratios earned by investors who are able to trade at theoretical prices, as computed over the various values of  $\eta_V$ . Clearly, even small values of  $\eta_V$  lead to economically significant expected returns and Sharpe ratios. For example,  $\eta_V = 0.011$  gives rise to Sharpe ratios ranging from 3 to almost 5 for a straddle issuer.

Figure 3 first and foremost serves as a way to interpret the volatility premium parameter  $\eta_V$ . Clearly, relatively small values of this parameter translate into significant gains to options issuers. This raises a number of questions. First, one may ask how the parameter value of 0.011 can be empirically validated. In Eraker (2004) this parameter value was identified primarily by fitting the model parameters to options prices and time series dynamics for the underlying asset. This procedure ensures that the theoretical model fit the observed data. It does not guarantee that the estimated  $\eta_V$  is economically reasonable.

It is reasonable to consider to whether equilibrium models can conceivably generate risk premiums and market risk prices consistent with those implicit in options prices. First off, equilibrium models based on CRRA preferences *cannot* explain the large premia because CRRA preferences imply that only shocks to consumption are priced. It is possible to generate a non-zero premium in models where the investor has utility of wealth. Perhaps most interesting is the recent literature on long run risk models. Bansal and Yaron (2004) show that their basic model generates non-zero market price of volatility risk. The premium is increasing in the level of risk aversion, and also as a function of the speed of mean reversion of volatility. The intuition here is that if a shock has a more persistent impact on the economy, it carries a higher reward to risk ratio. Eraker and Shaliastovich (2008) study a generalized version of the Bansal–Yaron model for option pricing, and show that the premium for both volatility and jump risks can be significant.



**Fig. 3** Expected returns (*top*) and Sharpe ratios (*bottom*) buying straddles at model prices as a function of  $\eta_V$ . *Left* hand graphs are computed assuming a zero equity premium, and *right* hand assumes an equity premium of 7.5%

## 5 Conclusion

This paper presents an alternative interpretation of pricing errors—the difference between market prices and prices obtained from a theoretical pricing model. We argue that if the market is mis-pricing assets, it may not be correct to dismiss a model because it has large pricing errors. Rather, if the model provides an objective, theoretically sound view of the expected payoff from holding the asset, positive or negative pricing errors may imply positive trading profits. In this situation, the paper argues, a statistical loss function based on the EUT is a useful way to compare competing pricing models.

The paper discusses the construction of mean-variance efficient portfolios based on the idea that expected returns are taken to equal the deviations between model and market prices, as well as the model implied risks. We apply this idea to data from options markets. The results show that an application of these strategies would have

produced reasonably high expected utility for a trader who is equipped with quadratic, mean-variance utility. In particular, the increase in Sharpe ratios (a sufficient statistic for the utility of a mean-variance investor) from following the hedging strategies is significant for the Heston SV model. We fail to find the increase in Sharpe ratio's for the hedging strategy statistically significant for the more elaborate SVCJ model. This is not necessarily because the hedging strategy does not work, or that the markets are really efficient, but rather that the pair-wise comparisons test for the Sharpe ratio's boil down to comparing the estimated means for the standardized, out-of-sample return distribution. Since the sample is relatively short, and the data are very noisy, the test does not have the power to reject the null.

The implications for future research are as follows. First, it would be desirable to conceivably develop tests that are more powerful than the pairwise comparisons tests used here. This is challenging because any test necessarily will need to be first-order dependent on the average rates return generated by trading strategies. Average rates of returns are notoriously imprecisely estimated. As such, there are three candidate ways to improve the test's power; increase the number of time-series data, increase the number of cross-sectional data, and/or seek to find better strategies for risk reduction of portfolio returns. For example, it is possible to consider broadly diversified portfolios that minimize systematic risk by identifying a set of common risk factors which can then be used to construct zero-beta portfolios as an alternative to the Markowitz approach in this paper.

The basic idea in this paper carries straightforwardly over to other asset markets. Consider first bond markets. There is a substantial literature on dynamic factor models for bond prices. Models based on latent factors typically fit prices very well, but fail to provide a structural interpretation of the changes in the yields. Structural models on the other hand are much more challenging to fit to the data. As such, it is reasonable to ask if the no-arbitrage models which provide almost perfect fit to observed yield curves do so because they are essentially over-fitted. To answer this question, take an comparably ill-fitting structural model which produces a clear equilibrium relationship between observed macro-economic variables and the yield curve. If such a model provide pricing errors which predict future bond returns, or returns on bond portfolios which are chosen optimally as suggested in this paper, the ranking of the two models could change.

Finally, the ideas in this paper apply to equity markets as well. The situation is somewhat more complex with equity returns, but the basic idea of EUT can still be applied in a number of ways. There are two complicating features of stock returns. First, if markets are really inefficient prices may take a very long time to revert to their equilibrium values. This is unlike equity options which represent short maturity bets on the probability distribution of the underlying stock. If stock prices deviate from their fundamentals, investors can at best hope that the markets correct themselves. Second, most models of equity prices are much less precise than are options and bond pricing models. This clearly make model uncertainty an issue difficult to overcome. Applications to stock and bond markets are interesting and challenging problems and will be pursued in future work.

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